

Seminar 10

week 10 (2 December 2024 – 6 December 2024)

- 1. Discussion of the assignment A6. The deadline of the assignment A6 is week 12 (15 - 19 December 2025). Please construct the typechecking tree proof for the following program using the ruled from Lecture 8:**

```
Prg= int v; Ref int a; v=10;new(a,22);  
    fork(wH(a,30);v=32;print(v);print(rH(a)));  
    print(v);print(rH(a))
```

Here is the starting point of the proof tree contruction. S is the rest of the program after int v. G is the type environment constructed during the typechecking. The constructed proof tree is the mathematical and logical proof of the given program correctness.

```
[]|- int v :void, [v:int]    [v:int] |- S:void, G  
-----  
[] |- intv; S ,, void , G
```

- 2. Discussion the implementation of the following problems (that can be problems for the final practical exam):**
 - a. implementation of synchronization mechanisms (lock, countdownlatch, semaphore, barrier, atomic operations),**
 - b. implementations of new IO operations (buffered writing/reading, writing to a file)**
 - c. implementation of some communications primitives (send/receive) between threads,**
 - d. adding procedures to the ToyLanguage.**
 - e. other constructions for the ToyLanguage**